Unit 5 - Managing Networks
Assignment 10: Upgrade a LAN Task 10

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 1

(U5P4) - Explain the functions of network management

Task description - *Research and explain six different network management functions, including user account management, fault management, checking network performance/traffic, user response times and network throughput/utilisation.*

Introduction

In this report, I will be researching the fundamentals of network management, then describing their purpose in a clear and intelligible way. So that I could give enough evidence that I have gathered sufficient knowledge and a comprehensive understanding on network management.

The network management functions that I will be explaining will be fault management, user account management and checking network performance/traffic, which would contain 3 more management functions which fall under performance management. Those being monitoring network user response times, network throughput and network line utilisation.

Fault management

One method of network management is through fault management. Which is done using a specific software that detects, isolates and resolves problems found within the network. Also, if installed correctly, has the capability to keep the network running at optimum level.

Essentially, a series of functions that detects network issues (such as scanning for viruses/Trojans, locating malfunctions or providing general diagnostics) are known as a fault management platform.

Additionally, some functions of a fault management platform allow the system to resolve issues without the need for human interaction. However, if the problem is too big for the system to resolve on its own, the system will alert the operator to take action on the problem before it starts affecting the network.

Where an operator would then access the fault management system and act on problems through a centralised console. This centralised console allows the operator to reconfigure or reboot the network from one place, which can make dealing with some problems easier.

To simply put it, there are certain situations when a fault management system requires network devices to be controlled remotely, meaning that a centralised console is necessary. Which brings on to the fact that there are two main types of fault management: active fault management and passive fault management.

Passive fault management collects alarms from devices via a Simple Network Management Protocol, or SNMP, when there is a problem with a device. So, with passive fault management, if a device fails completely an alarm may not be triggered and the problem will not be detected by the system.

Active fault management stops this issue by actively monitoring devices and determines if the device is functioning properly. If the device is not, the system will either attempt to find a solution if the problem is small, or it will trigger an alarm informing the operator that the device is not operational and needs to be fixed, as said in the beginning example.

User account management

Another method of a network management function would be account management. Which is control and order over different user accounts that will be found on a network, essentially controlling how they share information over the network to each other.

In order for user account management to work, at least one administrator account is needed so that there is a level of control over what the users can access and see within the network. These administrator accounts would be the only users that are able to keep control of the users and the network.

Not only can account management be used to administer current users, but can also be used to set up new accounts for users, manage old accounts and allocate rights to specific users. Which means that account management tools help administrators make changes to accounts and set up new accounts simply and easily.

When an administrator is setting up accounts, it is likely that the account will have a customary format. Such as: The date the user joined, forename, surname (to prevent multiple users having the same forename) and will also have to have to be password protected (to protect the data stored on that account).

However, In some networks passwords would need to be regularly changed, for security reasons. On these networks, an account management software would need to be configured to

do this. Meaning that administrators would have need to use account management tools to change the password if forgotten or if the security of account is compromised, due to multiple reasons (such as interception from an unknown source or leakage of sensitive information).

In larger networks, there will be multiple users accessing the network, most likely at the same time (i.e. the start of work hours). However, not all users will need to access the same applications or data. This can also be done using account management, which can allow the administrator to control what who gains access to what and when.

For instance, a user who has just started at a company would not be able to access the same information as a different user who has been there several years, unless their job requires that information.

For ease of management, users who need access to the similar applications and data tend to be grouped together as they have similar rights. It is this type of order that makes account management easier and setting up new accounts is quick and easy for new employees.

Likewise, a change to the user group will affect all users found in said group. For example, if some users in the group needed new a new software application, it would be made available to everyone in the group if the admin grants access, benefitting anyone who may need it in the future.

Checking network performance/traffic

Yet another method in network management is the control of network performance/traffic. Which is the monitoring of the consistency and speed of the data being sent over the network, to each of the users found within said network.

In most cases a network contains a lot of traffic, meaning that there is potential that the hardware containing the network may slow down due to multiple reasons. Meaning that network administrators need to be able to check the performance of the network, to prevent any future issues. Such as users being disconnected from the network due to the connection not being strong enough to sustain a stable connection or failing to access the network in the first place.

While performance management can help make sure that the network is running as well as possible, it can also find ways to improve the network and optimize the network if there is an issue. Which is extremely important for a network, due to the fact that improving the performance can bring beneficial outcomes to the entire network, maybe more so to specific sections of the network.

In order to manage the performance, administrators of the network will need a form of performance management (which will soon be mentioned) that can help track the network performance and traffic. With the specific methods, performance management tasks send

regular reports to the administrator monitoring the network. Reporting on different sections of network traffic and how they are currently performing.

So that the administrator can see a difference in performance, it will most likely include past reports to compare how each section is performing currently, to how it was performing the last time a report was sent. Allowing the administrator to determine whether the network can be improved or not based on the sections performance.

If the performance of a section is degrading, then the administrator may choose to reconfigure or replace devices to increase the performance of that section. Which the admin could then check on the reports on how effectively the network is currently working and if the network can or needs be improved.

In simple terms, the performance management can help the administrators determine on how well the network is functioning. Giving the admin information whether its performance is increasing or decreasing, in turn giving the admin the ability to check for improvement of performance, confirming that the change has worked and is benefiting the network.

To check changes in performance can be done in several methods, with each method detecting different types of aspects of a network that could potentially affect its performance. This is so it is easier for the administrator to pinpoint what is actually causing these performance changes. For instance, the most common methods are: measuring network throughput, user response times and measuring line utilisation.

User response times

Monitoring user response times is a simple and straightforward method of network management, where it will basically allow the admin to measure how quickly that the connectivity and performance to the users found in the network. To elaborate, user response time is measured on how quickly the network activity responds to the user's command.

For instance, if a large number of users are logging onto the network at one time, then the network response time may be a little slower as it is trying to complete multiple requests at once. Which the admin would then determine if the users are suffering performance issues, by how quickly the network responds to a request from the user.

In order for the admin to do this, there are multiple tools which allow someone to do so, one of the tools is a software called Nimsoft Monitoring Solution, or NMS, by the company Nimsoft. This can provide a numbers to measure and monitor how fast the response time is for the user.

Detecting user response time can also done by using commands such as traceroute, or “tracert” from the command prompt found in the Windows OS, or ping. The way that the software detects

the user response times can be done in various ways, such as: loading a web page, memory fetch or through a database query.

As mentioned before, the reason that this is being used is so that the admin can see how fast it is working for them. However, the administrator can also detect any errors with communication for the users using this type of software tool, realising that there is an error that needs to be addressed before it escalates into further problems.

Essentially, this type of network management exists so the administrator can see how the network performance is being sustained as more people start to use the network, hence why it also falls under the network performance management.

Measuring network throughput

Checking the network throughput is a simple but effective method of performance management within a network. Basically, network throughput is the measure the amount of data transmitted across the network. Which gives the admin the ability to ensure that the allocated data limits are not exceeded.

With this information, the admin can measure the traffic performance of the network, seen if the data is travelling across the network in an appropriate speed. While also checking if the data is too much for the network to handle, giving the admin the ability to understand any issues found with the data capacity and transfer rate.

The way that the speed and amount of data transmitted can be measured is through network throughput tools, such as: Iperf by The Iperf team, or BWPing by Oleg Dereventez. While there are plenty more software tools for this specific task, these are the most popular and well developed software.

These software tools are well known examples of measuring the bandwidth and response times between hosts in a network. However, with newer software components, network throughput can measure whether the device is improving the transmission of data across the network. Essnetially, these software tools can help administrators measure how quickly that data can travel through the network.

The way that an admin could test the network throughput is by downloading or uploading a large file over the network, monitoring the transmission speed through the communication device whether it can handle a sufficient amount of data. Basically, the reason that this function exists and carries out these methods is to see how fast information is getting form one part of the network to the other, which is why it falls under the network performance management.

Line utilisation

One method of measuring the network throughput, is through the line utilisation. While network throughput measures the amount of data moved from one place to another through any means of communication.

Line utilisation is measured through the cables specifically, which makes troubleshooting a much more accurate task for the administrator. To elaborate, line utilisation is the measurement of how much data is loaded onto the network cables that are currently being used, instead of the wide variety of data communication methods.

Which is why it is important for the administrator to measure, since it ensures that the amount of data being transmitted through the cable does not exceed the capability of what the cable line can handle.

The measurements for line utilisation are usually shown in a graph, which the administrator can make decisions off based on whether it would be best to upgrade the line or an alternative way to reduce the data transmission pressure on the cable. the amount of data on the line must not surpass the capability of the line.

For example, if the data on the cable exceeds the capability or the cable is underused, it would be causing performance issues within the network. So the administrator would troubleshoot the problem and use the line utilisation tools so the error can be reported and dealt with appropriately.

Essentially, this data is shown in a graph. Which the admin can use this information to make decisions on whether to upgrade the line or find alternative ways to reduce the pressure on the cable. Similarly done to measuring network throughput as a whole, since it is an effective way of troubleshooting performance issues with the throughput.

Conclusion

In conclusion, there are multiple ways in which an administrator of a network can manage the different functions found within said network. Those of which include monitoring user accounts, managing faults found in a network and checking the performance of the network, which is done through monitoring user response times, network throughput and line utilisation.

Below are the conclusions that I have discovered while researching through the network management methods, explaining is very simple detail of the knowledge I have gathered.

Firstly, fault management is the detection, isolation and resolving of problems found within the network by the administrator. Having fault management properly implemented can keep the network running at a standard level without the annoyance of the users accessing the network.

This is done by scanning for malicious software within the network such as viruses and malware, and could even detect and resolve any problem it may find within the network by itself through automation.

However, there are certain instances where the issue may be too significant for the fault management software. So an alert is sent to the admin, so it could be resolved by the administrator. Hence why a centralised console is essential for a network, also due to the fact the admin may need reset and reconfigure the network if needed.

Secondly, account management is when an administrator creates, deletes or modifies accounts found within the network for the rest of the users, as well as allocating access right for specific users. This is done by using tools to help manage user accounts and (when created for the users) requires essential information to identify the different users, such as: The joining date, forename, surname and password.

This is important so the network can differentiate who has access to which programs and other software in order to carry out their daily task. By giving groups of people different access rights, makes it much easier for the admin to make changes with the network. Such as installing new software and giving access to the specific users needed.

Checking network performance is essential because it is what makes the network run at an optimal rate, so that the users can access their intended programs and software to carry out daily tasks. The different ways of monitoring network performance is through user response times, network throughput and line utilisation.

User response times is the time measured on how fast the network responds to user activity. This is essential for the administrator to check when there are a large number of users logged into the network at a short time period. This is so the admin can check if every user is currently logged in and to fix any login problems, which may also be due to network throughput or utilisation limit.

Network throughput is essentially the measure of how much and how quickly data can and is travelling across the network, but by also making sure that the data limits aren’t actually exceeded. Which is basically used to measure the bandwidth and response to the host (mainly).

Line utilisation is essentially the amount of data that is loading onto the network cabling. Which is important to measure, because it ensures that the amount of data being transmitted through the cable does not exceed the capability of what the cable can actually handle.

Overall, managing a network is extremely important in order for the network to carry on working, since a network is never perfectly made, mistakes are bound to happen. Hence why all these types of network management functions exist, which is to cover all the possible mistakes that can be found in a network and report on problems which previously haven’t been encountered.